AI-BASED SMART BUS WINDOW

Circuit Connections and Physical Prototype Working

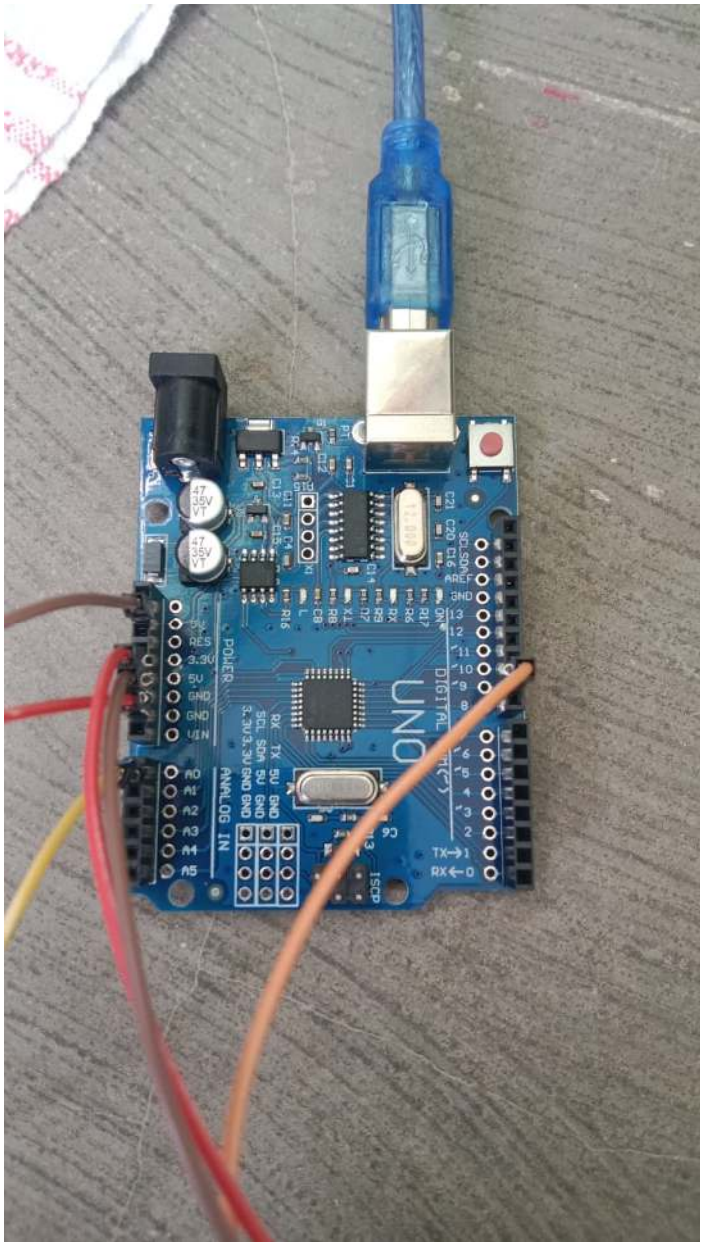

Figure 1. Arduino UNO used in the prototype with the passenger-control connections.

1. Prototype Overview

The Smart Bus Window prototype is an intelligent automatic window-control system designed for public transportation. The system combines automatic rain detection with manual passenger control. An Arduino UNO acts as the central controller and receives the rain-sensor input and passenger commands. An SG90 servo motor operates the miniature bus-window mechanism.

2. Overall System

The prototype has three control inputs:

- Rain sensor: detects water/rain and automatically commands the window to close.
- Passenger CLOSE input: allows the passenger to manually close the window.
- Passenger OPEN input: allows the passenger to manually open the window.

Overall signal flow:

RAIN SENSOR → ARDUINO UNO → SERVO MOTOR → BUS WINDOW
PASSENGER CLOSE (D2) → ARDUINO UNO → SERVO MOTOR
PASSENGER OPEN (D3) → ARDUINO UNO → SERVO MOTOR

3. Servo Motor Connections

The SG90 servo motor is responsible for physically moving the miniature bus-window mechanism. The signal wire is connected to Digital Pin 9. The red wire receives 5 V and the brown/black wire is connected to common ground.

Servo Wire	Function	Arduino UNO
Orange	Signal	Digital Pin 9
Red	VCC / Power	5V
Brown / Black	Ground	GND

4. Rain Sensor Connections

The rain sensor module is connected through its analog output (AO). This allows the Arduino to read a changing sensor value as the sensor surface becomes wet.

Rain Module Pin	Function	Arduino UNO
AO	Analog sensor output	A0
GND	Ground	GND
VCC	Power	5V

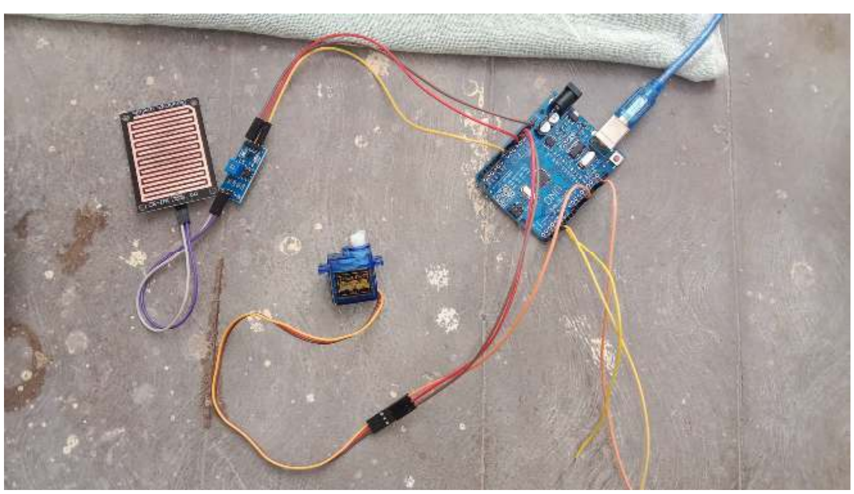

Figure 2. Complete physical prototype showing the rain sensor, Arduino UNO and SG90 servo.

5. Rain Detection Working

- Raindrops fall on the rain-sensor plate.
- The sensor output changes and is read by the Arduino through A0.
- The Arduino compares the reading with the programmed rain threshold.
- When rain is detected, the servo is commanded to move the window to the closed position.

6. Passenger CLOSE Input – Pin 2

The passenger CLOSE input is connected to Digital Pin 2. The input can be configured using the Arduino's internal pull-up resistor. When the CLOSE button is pressed, D2 is connected to GND and the Arduino commands the servo to close the window.

D2 —— CLOSE BUTTON —— COMMON GND

7. Passenger OPEN Input – Pin 3

The passenger OPEN input is connected to Digital Pin 3. When the OPEN button is pressed, D3 is connected to GND and the Arduino commands the servo to open the window.

D3 —— OPEN BUTTON —— COMMON GND

8. Common Ground Connection

Both passenger inputs share a common Arduino ground. Each button connects its respective digital pin to the common GND when pressed.

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 D2 ── CLOSE BUTTON ──┐
                      ├── GND
D3 ── OPEN BUTTON ────┘
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9. Control Logic

- Normal condition: no rain → window remains open.
- Rain detected → Arduino reads A0 → servo closes the window.
- Passenger presses CLOSE → D2 becomes active → servo closes the window.
- Passenger presses OPEN → D3 becomes active → servo opens the window.

10. Complete Connection Table

Component	Pin / Wire	Arduino UNO
Rain Sensor	AO	A0
Rain Sensor	GND	GND
Rain Sensor	VCC	5V
Servo Motor	Signal (Orange)	D9
Servo Motor	VCC (Red)	5V
Servo Motor	GND (Brown/Black)	GND
Passenger CLOSE	Button input	D2
Passenger OPEN	Button input	D3
Both buttons	Common ground	GND

11. Step-by-Step Working

Normal condition: When no rain is detected, the window remains open.

Rain detection: Water falls on the rain-sensor plate and the sensor reading changes.

Arduino processing: The Arduino reads the sensor value through A0 and checks the programmed threshold.

Automatic closure: When rain is detected, D9 sends the servo control signal and the window closes.

Manual CLOSE: The passenger presses the CLOSE input connected to D2 to close the window.

Manual OPEN: The passenger presses the OPEN input connected to D3 to open the window.

12. Key Features of the Prototype

- Automatic rain-based window closure.
- Manual passenger control for opening and closing.
- Servo-based physical window movement.
- Arduino-based sensing and control.
- Miniature demonstration of a smart public-transport window system.

13. Testing and Safety Notes

- Apply water only to the rain-sensor plate and keep the Arduino and other electronics dry.
- If the servo causes Arduino resets or behaves erratically, use a separate regulated 5 V supply for the servo and connect its GND to Arduino GND.
- Adjust the rain-sensor potentiometer and software threshold according to the actual sensor readings.
- Check the servo open and closed angles before permanently attaching it to the bus-window mechanism.

14. Prototype Summary

The developed prototype demonstrates an automated bus-window mechanism that responds to rainfall while also allowing passenger control. The Arduino UNO serves as the central controller, the rain sensor provides environmental input, and the SG90 servo provides the physical window movement. This combination demonstrates the core functionality of the proposed smart-bus window concept.